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Introduction:- I am prakhar kaushik currently pursuing a diploma in programming under a bs degree online and also pursuing b.tech from aktu university in electronics and communication engineering.

Description

In this project I have made a web app quizmaster in which there are two types of users. One is an admin who creates subject,chapter,quizzes and questions for students who can appear for the quiz and test their aptitude in each subject and also see their scores at the end of quiz.

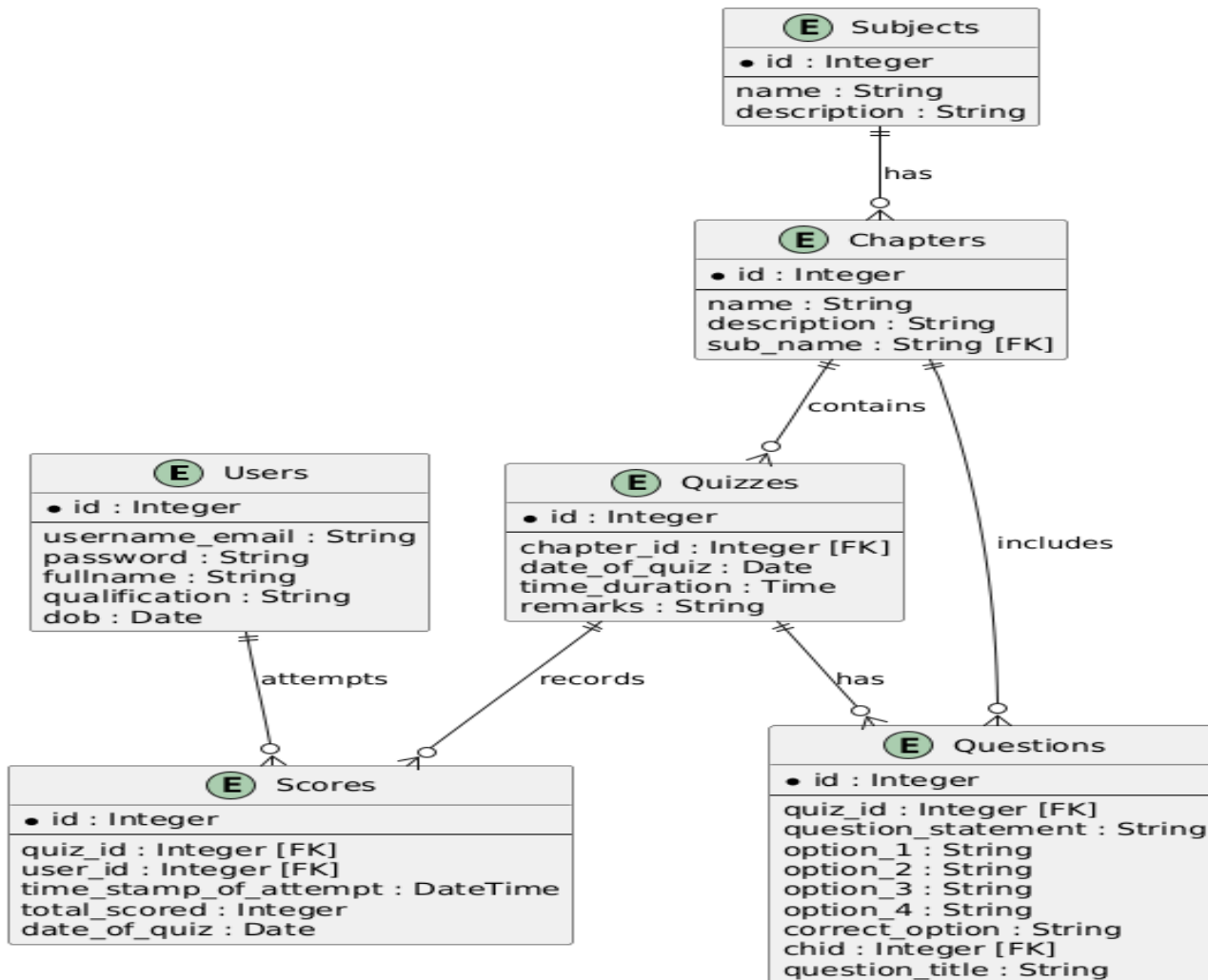
Technologies used

- ☐ Flask:- It creates an app environment in which we create our routes for redirecting urls and also provide basic layout for web apps.
- ☐ Render_template:- It is widely used for matching the html file to given url commands given by specified action created by certain tags in html web page.
- ☐ Request:- In this form we can take input from the input tag inside the html web page.
- ☐ Session:- It will provide a perfect encryption type for a certain time interval so that other outside interference do not affect the web page while processing and also authentication of users.
- ☐ Sqlite3:- It will provide us to create database and use function while doing query execution.While there are many useful cases to provide relation between data sets implemented.While it also allows us to create tables,insert data in different tables,provide to select the specific data from tables,it helps to make web application to be compatible with CRUD platform.
- ☐ Jinja2:- While rendering html web pages we have specific data passed through html web pages so it will show dynamic responses to each specified route.
- ☐ Css:- basics css is used for making html web pages look good and decorative so the user interface is somehow better than normal raw html web page.
- ☐ Matplotlib:- It is used for showing data pictorially and in a more eye-catching way it provides summary of scores and plot in chart form.
- ☐ Io:- In this I have used ByteIO() in it will provide buffer storage for temporary use and also process the binary data for seek purpose.
- ☐ Encoding:- In this project there is encoding done for converting matplotlib graphs in such a form that it will be used in a manner which can provide encoded data for temporary registry purposes.
- ☐ Decoding:-In this encoded data is then processed and converted into utf-8 form so that png image can be saved in a smooth process without losing its binary data.

DB Schema Design

For interaction in practical bases we ensure our data is collectively stored in such a way that it can easily be accessed and used in the form of some relation in data such that it can be utilized in usage purpose.

The ER-DIAGRAM IS shown below:-



- Details of the tables is given by the given points shown under :-
 - Subjects:- It has subject names which all subject names stored with their unique id.
 - Chapter:- It has chapter id, name and subject name.
 - Quizzes:-The structure includes the following elements: ID, date of the quiz, time duration, and remarks.
 - Users:- It has username,password,full name , qualification and dob as entry used to carry basic information of users.
 - Scores:- It has quiz id,user id,time_stamp_of _attempt,total score, and date of quiz attempted.It stores overall performance of users.
 - Question:- In this all questions related to the subject and specified chapter are stored in a table with all options with correct option and question title in it.
- Relations between tables:-
 - A subject has many chapters (1:M).
 - A chapter has many quizzes and questions (1:M).
 - A quiz has many questions and scores (1:M).
 - A user has many scores (1:M).

Architecture and Features

In this there is explanation of architecture and feature is addressed :-

- Login page:- In this user and admin enter their entries by their specified username and password,if user does not exist then user can easily make a new id.
- Landing page :- It has two sections, one for user and admin.
 - Admin :- it has many sections which allows admin to create and delete edit the specified objects in a web application.
 - Admin has access to creating questions,quiz,subject and chapters.
 - Users :- It has in display all the current quizzes made by admin. Users can attempt it any number of times.
- Quiz management:- In this all quizzes are managed and new quizzes can be added as well as new questions can be added,can be deleted only admin.
- Question attempt page:- It has a set of questions come one by one in a session so that score is updated in a form of table.
- Scores :- All the score of the quiz attempted by the user is shown in the dashboard.
- Summary:- It has two section one for user and one for admin:-
 - Admin :- In this performance all users are stored and displayed to admin to check performance of users.
 - User :- In these details the current user is shown on the dashboard.
- Search :- This option is given to both admin and user.By this admin or user can search the thing within the boundaries of their usage.

Video

https://drive.google.com/file/d/1MPweadMrua1NtccAlmK9ebBCA50uP_cW/view?usp=sharing